

Matrix-Kronecker-Product-Based Multichannel Linear Prediction for Speech Dereverberation

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Abstract—Existing KP-based methods for speech dereverberation are inherently limited to vectorized filter representations, which restrict their applicability to general multi-input multi-output (MIMO) configurations without structural reformulation. To address this limitation, we propose a matrix-Kronecker-product-based MIMO weighted prediction error method (MKP-MWPE), which introduces a direct matrix decomposition strategy that factorizes the high-dimensional MIMO prediction filter into two lower-dimensional matrix factors. Within the weighted-prediction-error (WPE) framework, we derive a rigorous optimization procedure for estimating these factors, establishing the first tractable formulation for matrix-form KP-based dereverberation. This formulation significantly reduces the number of free parameters while enforcing a compact structural representation of the reverberation process. Simulation results demonstrate the feasibility and effectiveness of the proposed method for speech dereverberation, although the coupling between the matrix factors leads to a more challenging optimization landscape that requires iterative refinement.

Index Terms—Matrix Kronecker product, WPE, dereverberation.

I. INTRODUCTION

Reverberation is a common phenomenon in enclosed acoustic environments, where speech signals reflect off walls, ceilings, and other surfaces before reaching the microphones [1]–[4]. These reflections, particularly late reflections that constitute reverberation, degrade both speech intelligibility and perceived quality, posing significant challenges for speech communication systems and intelligent speech interaction applications [5]–[8]. Effective dereverberation is therefore essential for improving performance in applications such as automatic speech recognition, teleconferencing, and hearing aids [9]–[12].

Among various dereverberation methods developed in the literature, the weighted-prediction-error (WPE) approach has demonstrated strong effectiveness [13], [14]. To exploit spatial information from multiple microphones, the multi-input multi-output (MIMO) extension of WPE is typically adopted. However, this improvement in performance comes at the cost of significant parameter redundancy. A standard MIMO WPE prediction filter is represented by a high-dimensional matrix whose number of parameters scales quadratically with the number of microphones and linearly with the prediction order [15]–[17]. This rapid growth in dimensionality imposes substantial computational and memory burdens, making real-time or resource-constrained deployment challenging.

One effective approach for reducing the computational complexity of speech dereverberation algorithms is the use of Kronecker product (KP) decomposition, which serves as a powerful tool for mitigating parameter redundancy in multichannel linear prediction (MCLP) filters [18]–[23]. However, existing KP-based WPE methods are

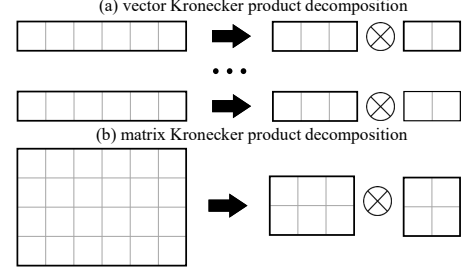


Fig. 1. Conceptual comparison of vector-form and matrix-form Kronecker product decompositions.

fundamentally restricted to vector-form filter representations. These approaches typically reformulate the multichannel system as a set of multi-input single-output (MISO) sub-problems or reshape the MIMO prediction filter into a long vector prior to applying KP decomposition. Consequently, they can only capture correlations within vectorized subspaces and fail to exploit the inherent matrix-level structure of true MIMO prediction filters.

This limitation is conceptually illustrated in Fig. 1. As shown in Fig. 1(a), conventional KP-based WPE methods apply KP decomposition to vectorized prediction filters, effectively decomposing the MIMO system into independent MISO components [19], [24]. Although this formulation reduces redundancy within each output channel, it necessarily discards the native matrix structure and ignores inter-channel coupling. In contrast, Fig. 1(b) depicts the proposed matrix-form KP decomposition, where the MIMO prediction filter is directly factorized as the KP of two matrix-form sub-filters. This formulation preserves the matrix-level interactions across both spatial and temporal dimensions.

Motivated by this observation, we propose a matrix-form Kronecker product decomposition for MIMO WPE (MKP-MWPE). In contrast to existing KP-based methods that rely on vectorized filters, the proposed formulation directly factorizes the MIMO prediction matrix as the KP of two smaller matrix-valued filters. This extension is non-trivial, as standard vector-based KP derivations cannot be directly applied to matrix-valued factors. We then develop a rigorous optimization framework for estimating the coupled matrix sub-filters within the MKP-MWPE formulation, leading to an efficient alternating minimization algorithm that iteratively solves the resulting optimization problem. Simulation results validate the feasibility of the proposed approach.

II. SIGNAL MODEL AND PROBLEM FORMULATION

Consider a multichannel acoustic acquisition system in which an array of M microphones records a target speech signal in a reverberant

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and noisy environment. The signal observed at the m th microphone is given by

$$y_m(t) = h_m(t) * s(t) + v_m(t), \quad m = 1, 2, \dots, M, \quad (1)$$

where $s(t)$ denotes the clean target source signal, $h_m(t)$ represents the acoustic impulse response between the source and the m th microphone, $v_m(t)$ corresponds to the additive noise at the m th microphone, and $*$ denotes linear convolution. It is assumed that the reverberant speech components, i.e., $h_m(t) * s(t)$ for $m = 1, 2, \dots, M$, are spatially coherent across microphones, while the noise is uncorrelated with the speech source. All signals are assumed to be zero-mean, real-valued, and broadband.

Applying the short-time Fourier transform (STFT) to both sides of (1), and considering the general case where the STFT window length is not necessarily longer than the acoustic impulse responses, the time-domain signals can be represented in the STFT domain using the convolutive transfer function (CTF) model as [14], [25]

$$Y_m(n, k) = \sum_{l=0}^{L_g-1} H_m(l, k) S(n-l, k) + V_m(n, k), \quad (2)$$

$$m = 1, 2, \dots, M,$$

where $H_m(l, k)$, of length L_g , denotes the STFT-domain representation of the room impulse response, $Y_m(n, k)$, $S(n, k)$, and $V_m(n, k)$ are, respectively, the STFT coefficients of the microphone signal, clean speech, and noise, and the indices n and k represent the time frame and frequency bin. By stacking the M microphone signals into a single vector, the multichannel STFT-domain model can be written compactly as

$$\mathbf{y}(n, k) = [Y_1(n, k) \ Y_2(n, k) \ \dots \ Y_M(n, k)]^T, \quad (3)$$

$$= \sum_{l=0}^{L_g-1} \mathbf{h}(l, k) S(n-l, k) + \mathbf{v}(n, k),$$

where $\mathbf{h}(l, k) = [H_1(l, k) \ H_2(l, k) \ \dots \ H_M(l, k)]^T \in \mathbb{C}^{M \times 1}$ is the vector of acoustic transfer functions for all microphones, T denotes the transpose, and $\mathbf{v}(n, k) \in \mathbb{C}^{M \times 1}$ denotes the multichannel noise vector.

The multichannel linear prediction (MCLP) approach performs dereverberation by estimating the late reverberation components from past observations and suppressing them from the current microphone signals, thereby recovering the direct-path and early reflection components. This process can be expressed as

$$\mathbf{d}(n, k) = \mathbf{y}(n, k) - \mathbf{G}^H(k) \bar{\mathbf{y}}(n - \Delta, k), \quad (4)$$

where the superscript H denotes the conjugate transpose, $\mathbf{d}(n, k) \in \mathbb{C}^{M \times 1}$ represents the estimated direct-path-plus-early-reflection speech components, $\mathbf{G}(k) \in \mathbb{C}^{ML \times M}$ is the prediction filter, Δ denotes the prediction delay, and the stacked observation vector is defined as

$$\bar{\mathbf{y}}(n - \Delta, k) = [\mathbf{y}^T(n - \Delta, k) \ \mathbf{y}^T(n - \Delta - 1, k) \ \dots \ \mathbf{y}^T(n - \Delta - L + 1, k)]^T \in \mathbb{C}^{ML \times 1}. \quad (5)$$

Based on the signal model in (4), dereverberation reduces to estimating the prediction filter $\mathbf{G}(k)$. In the WPE framework [14], the desired speech component in the STFT domain is modeled as a zero-mean complex Gaussian random variable with time-varying variance. Accordingly, the following cost function is formulated:

$$\mathcal{L}(\mathbf{G}(k)) = \sum_{n=1}^N \frac{\|\mathbf{y}(n, k) - \mathbf{G}^H(k) \bar{\mathbf{y}}(n - \Delta, k)\|_2^2}{\lambda(n, k)}, \quad (6)$$

where N denotes the total number of time frames and $\|\cdot\|_2$ represents the ℓ_2 -norm. The time-varying variance is estimated as

$$\lambda(n, k) = \frac{1}{M} \|\mathbf{d}(n, k)\|_2^2. \quad (7)$$

Minimizing the above cost function leads to the following closed-form solution for the prediction filter [14]:

$$\mathbf{G}(k) = \mathbf{R}^{-1} \mathbf{P}, \quad (8)$$

where

$$\mathbf{R} = \sum_{n=1}^N \frac{\bar{\mathbf{y}}(n - \Delta, k) \bar{\mathbf{y}}^H(n - \Delta, k)}{\lambda(n, k)} \in \mathbb{C}^{ML \times ML}, \quad (9)$$

$$\mathbf{P} = \sum_{n=1}^N \frac{\bar{\mathbf{y}}(n - \Delta, k) \mathbf{y}^H(n, k)}{\lambda(n, k)} \in \mathbb{C}^{ML \times M} \quad (10)$$

represent the weighted spatio-temporal covariance matrices.

In practical implementations, the prediction filter $\mathbf{G}(k)$ and the variance $\lambda(n, k)$ are updated alternately in an iterative manner, starting from an initial estimate, until convergence to a locally optimal solution is reached. As can be seen from the above formulation, the standard MIMO WPE prediction filter $\mathbf{G}(k) \in \mathbb{C}^{ML \times M}$ contains $M^2 L$ complex-valued parameters per frequency bin. This parameterization grows quadratically with the number of microphones and linearly with the prediction order, leading to substantial computational and memory demands in practical applications.

Although Kronecker product decompositions have been used to reduce parameter redundancy in WPE-based dereverberation, existing methods are still fundamentally limited to vectorized filter representations. In particular, these approaches often reformulate the multichannel system as a set of independent MISO sub-problems, rather than leveraging the full structure of the MIMO system. This vectorization not only masks the underlying matrix-level interactions between all input-output channel pairs, but also limits straightforward extension to more general MIMO architectures.

III. MATRIX KRONECKER PRODUCT DECOMPOSITION BASED WPE

To address the aforementioned limitations, we propose a Kronecker product decomposition framework that operates directly on matrix-form MCLP filters. To this end, we express the filter $\mathbf{G}(k)$ as a Kronecker product of two smaller filter matrices:

$$\mathbf{G}(k) = \mathbf{G}_1(k) \otimes \mathbf{G}_2(k), \quad (11)$$

where $\otimes$ denotes the Kronecker product, $\mathbf{G}_1(k) \in \mathbb{C}^{L_{11} \times L_{12}}$ and $\mathbf{G}_2(k) \in \mathbb{C}^{L_{21} \times L_{22}}$ are lower-dimensional matrix filters satisfying $L_{11} L_{21} = ML$ and $L_{12} L_{22} = M$. Note that both $\mathbf{G}_1(k)$ and $\mathbf{G}_2(k)$ are significantly smaller than $\mathbf{G}(k)$; as a result, this reformulation reduces the number of parameters to be estimated and has the potential to lower computational complexity. Since the filters are estimated independently for each frequency bin, the frequency index k is omitted in the subsequent derivations for clarity.

The dereverberation process can then be expressed as

$$\mathbf{d}(n) = \mathbf{y}(n) - (\mathbf{G}_1^H \otimes \mathbf{G}_2^H) \bar{\mathbf{y}}(n - \Delta). \quad (12)$$

By the properties of the Kronecker product, the following relationship holds:

$$(\mathbf{G}_1^H \otimes \mathbf{G}_2^H) \bar{\mathbf{y}}(n - \Delta) = \text{vec}(\mathbf{G}_2^H \bar{\mathbf{Y}}(n - \Delta) \mathbf{G}_1^*), \quad (13)$$

where $\text{vec}(\cdot)$ denotes the vectorization operator that stacks the columns of a matrix into a single vector, and $*$ denotes complex conjugation. The matrix:

$$\bar{\mathbf{Y}}(n - \Delta) = \text{ivec}_{L_{21}, L_{11}}(\bar{\mathbf{y}}(n - \Delta)) \in \mathbb{C}^{L_{21} \times L_{11}} \quad (14)$$

is obtained by applying the inverse vectorization operator $\text{ivec}_{p,q}(\cdot)$, matrix in a column-wise fashion. Substituting (13) into (12) yields the dereverberated:

$$\mathbf{d}(n) = \mathbf{y}(n) - \text{vec}(\mathbf{G}_2^H \bar{\mathbf{Y}}(n - \Delta) \mathbf{G}_1^*). \quad (15)$$

With the above formulation, the dereverberation problem reduces to estimating two sub-filter matrices, $\mathbf{G}_1$ and $\mathbf{G}_2$. Assuming again that the desired speech signal follows a zero-mean complex Gaussian distribution with time-varying variance, the cost function for estimating these two sub-filter matrices can be formulated as

$$\mathcal{L}(\mathbf{G}_1, \mathbf{G}_2) = \sum_{n=1}^N \frac{\|\mathbf{y}(n) - \text{vec}(\mathbf{G}_2^H \bar{\mathbf{Y}}(n - \Delta) \mathbf{G}_1^*)\|_2^2}{\lambda(n)}. \quad (16)$$

Since the two sub-filters are interdependent, jointly estimating them in a single step is nontrivial. Therefore, we adopt a two-stage optimization procedure in which each sub-filter is updated alternately. We first estimate $\mathbf{G}_1$ while keeping $\mathbf{G}_2$ fixed. For notational convenience, we define

$$\bar{\mathbf{Y}}_{\mathbf{G}_2}(n - \Delta) = \mathbf{G}_2^H \bar{\mathbf{Y}}(n - \Delta) \in \mathbb{C}^{L_{22} \times L_{11}}. \quad (17)$$

Furthermore, we denote the reshaped version of $\mathbf{y}(n)$ as

$$\bar{\mathbf{Y}}(n) = \text{ivec}_{L_{22}, L_{12}}(\mathbf{y}(n)) \in \mathbb{C}^{L_{22} \times L_{12}}. \quad (18)$$

With these definitions, the squared ℓ_2 -norm involved in the cost function can be equivalently expressed using the Frobenius norm as

$$\mathcal{L}(\mathbf{G}_1 | \mathbf{G}_2) = \sum_{n=1}^N \frac{\|\bar{\mathbf{Y}}(n) - \bar{\mathbf{Y}}_{\mathbf{G}_2}(n - \Delta) \mathbf{G}_1^*\|_F^2}{\lambda(n)}. \quad (19)$$

Taking the derivative with respect to $\mathbf{G}_1$ using Wirtinger calculus and setting it to zero yields the following solution:

$$\mathbf{G}_1 = \mathbf{R}_{\mathbf{G}_1}^{-1} \mathbf{P}_{\mathbf{G}_1}, \quad (20)$$

where

$$\mathbf{R}_{\mathbf{G}_1} = \sum_{n=1}^N \frac{\bar{\mathbf{Y}}_{\mathbf{G}_2}^T(n - \Delta) \bar{\mathbf{Y}}_{\mathbf{G}_2}^*(n - \Delta)}{\lambda(n)} \in \mathbb{C}^{L_{11} \times L_{11}}, \quad (21)$$

$$\mathbf{P}_{\mathbf{G}_1} = \sum_{n=1}^N \frac{\bar{\mathbf{Y}}_{\mathbf{G}_2}^T(n - \Delta) \bar{\mathbf{Y}}^*(n)}{\lambda(n)} \in \mathbb{C}^{L_{11} \times L_{12}}. \quad (22)$$

Similarly, when estimating $\mathbf{G}_2$, $\mathbf{G}_1$ remains fixed. In this step, the cost function for $\mathbf{G}_2$ is expressed as

$$\mathcal{L}(\mathbf{G}_2 | \mathbf{G}_1) = \sum_{n=1}^N \frac{\|\bar{\mathbf{Y}}(n) - \mathbf{G}_2^H \bar{\mathbf{Y}}_{\mathbf{G}_1}(n - \Delta)\|_F^2}{\lambda(n)}, \quad (23)$$

where

$$\bar{\mathbf{Y}}_{\mathbf{G}_1}(n - \Delta) = \bar{\mathbf{Y}}(n - \Delta) \mathbf{G}_1^* \in \mathbb{C}^{L_{21} \times L_{12}}. \quad (24)$$

By minimizing this objective, we obtain the optimal solution for $\mathbf{G}_2$ as [14]

$$\mathbf{G}_2 = \mathbf{R}_{\mathbf{G}_2}^{-1} \mathbf{P}_{\mathbf{G}_2}, \quad (25)$$

Algorithm 1 The MWPE-MKP method.

Input: $M, L, \Delta, L_{11}, L_{12}, L_{21}, L_{22}, \mathcal{I}, \mathbf{y}(n, k), \forall n, k$
Output: $\mathbf{G}_1, \mathbf{G}_2, \mathbf{d}(n)$

- 1: **for** $i = 1, 2, \dots, \mathcal{I}$ **do**
- 2: Compute the dereverberated signal and its variance
- 3: $\mathbf{d}(n) = \mathbf{y}(n) - (\mathbf{G}_1^H \otimes \mathbf{G}_2^H) \bar{\mathbf{y}}(n - \Delta), \forall n, k$
- 4: $\lambda(n) = \frac{1}{M} \|\mathbf{d}(n)\|_2^2, \forall n, k$
- 5: Calculate $\bar{\mathbf{Y}}_{\mathbf{G}_2}(n - \Delta) = \mathbf{G}_2^H \bar{\mathbf{Y}}(n - \Delta), \forall n, k$
- 6: Calculate $\mathbf{R}_{\mathbf{G}_1}$ and $\mathbf{P}_{\mathbf{G}_1}$ according to (21) and (22)
- 7: Update the prediction sub-filter
- 8: $\mathbf{G}_1 = \mathbf{R}_{\mathbf{G}_1}^{-1} \mathbf{P}_{\mathbf{G}_1}, \forall k$
- 9: Calculate $\bar{\mathbf{Y}}_{\mathbf{G}_1}(n - \Delta) = \bar{\mathbf{Y}}(n - \Delta) \mathbf{G}_1^*, \forall n, k$
- 10: Calculate $\mathbf{R}_{\mathbf{G}_2}$ and $\mathbf{P}_{\mathbf{G}_2}$ according to (26) and (27)
- 11: Update the prediction sub-filter
- 12: $\mathbf{G}_2 = \mathbf{R}_{\mathbf{G}_2}^{-1} \mathbf{P}_{\mathbf{G}_2}, \forall k$
- 13: **end for**

TABLE I
KRONECKER PRODUCT DECOMPOSITION CONFIGURATIONS FOR MWPE-MKP.

method	$\mathbf{G}_1$		$\mathbf{G}_2$	
MWPE-MKP-1	$L_{11} = 15$	$L_{12} = 3$	$L_{21} = 12$	$L_{22} = 3$
MWPE-MKP-2	$L_{11} = 20$	$L_{12} = 3$	$L_{21} = 9$	$L_{22} = 3$
MWPE-MKP-3	$L_{11} = 30$	$L_{12} = 3$	$L_{21} = 6$	$L_{22} = 3$
MWPE-MKP-4	$L_{11} = 20$	$L_{12} = 9$	$L_{21} = 9$	$L_{22} = 1$
MWPE-MKP-5	$L_{11} = 30$	$L_{12} = 9$	$L_{21} = 6$	$L_{22} = 1$

where

$$\mathbf{R}_{\mathbf{G}_2} = \sum_{n=1}^N \frac{\bar{\mathbf{Y}}_{\mathbf{G}_1}(n - \Delta) \bar{\mathbf{Y}}_{\mathbf{G}_1}^H(n - \Delta)}{\lambda(n)} \in \mathbb{C}^{L_{21} \times L_{21}}, \quad (26)$$

$$\mathbf{P}_{\mathbf{G}_2} = \sum_{n=1}^N \frac{\bar{\mathbf{Y}}_{\mathbf{G}_1}(n - \Delta) \bar{\mathbf{Y}}^H(n)}{\lambda(n)} \in \mathbb{C}^{L_{21} \times L_{22}}. \quad (27)$$

We iteratively update $\mathbf{G}_1$ and $\mathbf{G}_2$ for $\mathcal{I}$ iterations until convergence is achieved. After convergence, the final dereverberation filters are obtained and the clean speech signal $\mathbf{d}(n)$ is recovered using (12). The original multi-input multi-output WPE method is referred to as MWPE, while the proposed approach, which incorporates a matrix Kronecker product decomposition within the MIMO WPE framework, is termed MKP-MWPE. The overall procedure of the proposed MWPE-MKP method is summarized in Algorithm 1.

IV. SIMULATIONS

A. Simulation Setup

The clean speech signals used in the simulations are taken from the TIMIT dataset [26], recorded at a sampling rate of 16 kHz. The acoustic channel impulse responses between the source and microphones are generated using the image-source method [27]. We consider a room of size $6 \times 5 \times 4$ m with six walls sharing identical reflection coefficients, corresponding to a reverberation time T_{60} of 600 ms. A uniform linear array consisting of 9 omnidirectional microphones is employed, with an inter-microphone spacing of 0.02 m. The sound source is placed 2 m away from the array in the endfire direction. The observed microphone signals are obtained by convolving the clean speech signal with the corresponding room impulse responses. Dereverberation is performed in the STFT domain, where the observed signals are segmented into overlapping frames of 512 samples each.

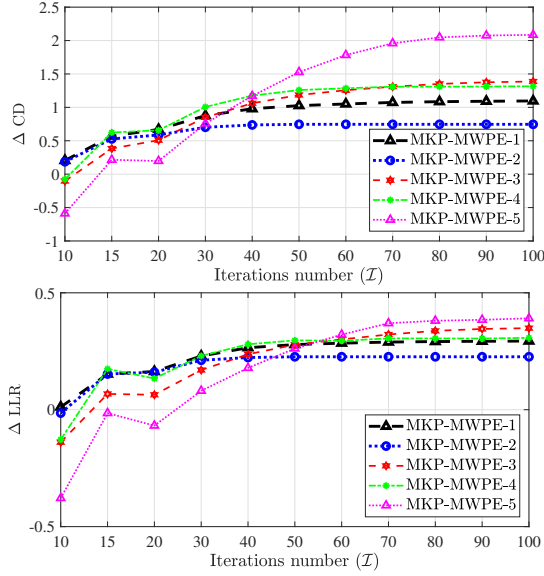


Fig. 2. Improvements in CD and LLR achieved by the proposed MKP-MWPE method under different decomposition configurations.

A Kaiser window with 75% overlap is used for STFT analysis and synthesis.

To evaluate dereverberation performance, we adopt cepstral distance (CD) [28], frequency-weighted segmental signal-to-noise ratio (FWSNR) [29], perceptual evaluation of speech quality (PESQ) [29], and short-time objective intelligibility (STOI) [30] as performance metrics. The reference signal for all metrics is chosen as the direct-path component plus early reflections within 40 ms. The prediction delay is set to $\Delta = 5$, and the prediction order is $L = 20$. For the MKP-MWPE method, five decomposition configurations are considered, as summarized in Table I.

B. Results and Analysis

The proposed MKP-MWPE method is evaluated using Δ PESQ, Δ FWSNR, Δ CD, and Δ LLR, where Δ denotes the improvement relative to the corresponding metric of the observed signal. Figures 2 and 3 illustrate the performance evolution over 100 iterations for five different decomposition configurations. All configurations show stable, nearly monotonic improvements, confirming the effectiveness of the proposed alternating optimization scheme.

The simulation results show that the dereverberation performance of MKP-MWPE is strongly dependent on the chosen matrix decomposition configuration. Among the five tested settings, MKP-MWPE-5 consistently yields the largest improvements across all metrics, whereas configurations such as MKP-MWPE-2 achieve noticeably smaller gains. This highlights a key characteristic of the proposed method: the effectiveness of the matrix-form Kronecker product decomposition is highly influenced by how the original MIMO prediction filter is factorized. In particular, decompositions that distribute the degrees of freedom more evenly between the two sub-filters tend to better preserve the spatio-temporal structure of the original MIMO filter, resulting in improved dereverberation performance.

Regarding absolute performance, the proposed MKP-MWPE method does not yet outperform the conventional MIMO WPE algorithm. This is expected, since the method introduces a structured low-rank constraint on the prediction filter, which inherently reduces the solution space. Nevertheless, the proposed framework operates directly

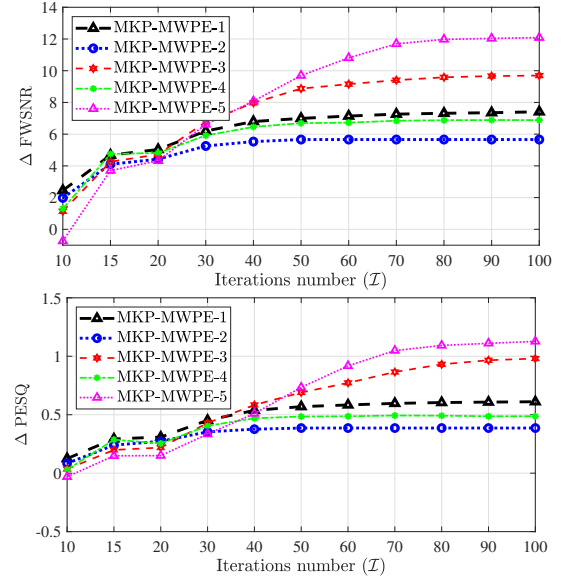


Fig. 3. Improvement in FWSNR and PESQ achieved by the proposed MKP-MWPE method under different decomposition configurations.

on the native MIMO matrix-form filter, in contrast to existing KP-based approaches that rely on vectorized or MISO reformulations. The results indicate that, even under such structural constraints, MKP-MWPE still achieves meaningful dereverberation gains, thereby validating the feasibility of matrix-form Kronecker product decomposition for structured MIMO linear prediction. Another observation is that MKP-MWPE requires more iterations to converge than conventional WPE, due to the strong coupling between the matrix factors. However, the algorithm converges reliably and consistently yields stable performance improvements across all tested configurations.

Overall, the simulation results demonstrate that the proposed MKP-MWPE framework offers an effective and mathematically principled approach to structured MIMO dereverberation. Although its performance does not yet match that of unconstrained MIMO WPE, it clearly highlights the potential of matrix-form Kronecker product decomposition for reducing model complexity while preserving essential dereverberation capability.

Future work will focus on accelerating convergence, reducing the number of required iterations, and refining the matrix decomposition strategy to further enhance both computational efficiency and dereverberation performance.

V. CONCLUSIONS

This paper presented MKP-MWPE, a novel multichannel speech dereverberation framework that incorporates matrix-form Kronecker product decomposition into the MIMO WPE formulation. In contrast to existing KP-based approaches that rely on vectorized filters, the proposed method directly factorizes the MIMO prediction matrix into two smaller matrix filters, establishing a first solvable matrix-level Kronecker decomposition within the WPE framework and substantially reducing the number of free parameters. Simulation results showed that MKP-MWPE is able to achieve consistent dereverberation improvements, although its performance is strongly influenced by the chosen matrix decomposition configuration; in particular, more balanced factorizations tend to yield better results. While the proposed method does not surpass conventional MIMO WPE in terms of absolute performance, it demonstrates the feasibility and effectiveness of structured matrix-form modeling for multichannel dereverberation.

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